

Assignment 1 - Discrete Mathematics (Monsoon 2025)

Answer Key

“Each problem that I solved became a rule which served afterwards to solve other problems.”
- by René Descartes

Submission deadline: 11:59 PM on 9th September, 2025.

Maximum marks: 60

Instructions. Consider the following while preparing the answers for the given questions.

1. Kindly go over the academic dishonesty policy of IIIT Delhi given on [this link](#). Any deviation from this policy will lead to the strictest possible punishment.
2. Each question is of 10 marks.
3. It is prohibited to use AI-related tools like chatGPT to solve the questions, unless explicitly mentioned. Any violation in this regard will be handled very strictly.
4. You are encouraged to write solutions in L^AT_EX. Anyone who submits homework in L^AT_EX (i.e., in the pdf format and not the source code) will get an additional 20% of their score in that homework.
5. While writing the solutions, you can refer to the topics, theorems, examples, etc., taught in the class. It is not allowed to refer to anything related to propositional and predictive logic not covered in the class.
6. Late submissions are not allowed.
7. If you discuss anything related to the solution of a question in this assignment with anyone, you must mention their names.

1 Problem Set

In this document, $\equiv$ means equivalence of propositions.

Question 1.1. 1. (6 marks). Let p, q, r be propositional variables. Write the truth tables of the following propositions.

(a) $\neg(p \oplus q) \implies \neg q \vee r$.

(b) $p \vee \neg r \iff q \wedge r$.

(c) $(p \oplus \neg q) \implies q \wedge \neg r$.

(d) $p \wedge (p \iff q) \implies \neg q$.

Answer hint: Use the precedence of logical operators.

2. (2 marks). Convert the following to predicates with quantifiers and then write the negations of these quantified predicates.

- (a) *There exists a student at IIIT Delhi who knows swimming and driving.*

Answer: Let x be a variable whose domain is all the students of IIIT Delhi, $P(x), Q(x)$ be the predicates that x knows swimming and x knows driving, respectively. Then, the statement given above can be logically expressed as $\exists x(P(x) \wedge Q(x))$. Now, on applying DeMorgan's laws of Quantifiers and propositional logic,

$$\neg \exists x(P(x) \wedge Q(x)) \equiv \forall x(\neg P(x) \vee \neg Q(x)).$$

This can be translated to English as: every student of IIIT Delhi either does not know swimming or does not know driving.

(b) Every fruit in the basket is either a mango or a strawberry.

Similar to the first part, the negation of this statement is as follows: there exists a fruit in the basket which is not a mango and not a strawberry.

3. (2 marks). Let x be a variable and $P(x)$ be a predicate. Explain why $\forall xP(x)$ and $\exists xP(x)$ are propositions?

Answer: For both $\forall xP(x)$ and $\exists xP(x)$, we can describe when they are true and when they are false. As predicates are declarative statements, we get that $\forall xP(x)$ and $\exists xP(x)$ are either true or false, but not both. Hence, these are propositions.

Question 1.2. Let p, q, r be propositions. State whether the following are true, false, or neither. Justify your answers.

1. $p \oplus \neg p \equiv p$.

Answer: This is false as the equivalence does not hold when we substitute F in p .

2. $\neg(p \vee q \vee r) \equiv \neg p \wedge \neg r \vee q$.

Answer: This is false. The equivalence does not hold if we substitute $p = T, q = T, r = T$.

3. $p \iff p \equiv \neg p \oplus \neg p$.

Answer: This is false. The equivalence does not hold for $p = T$.

4. $p \implies (p \wedge (p \vee p)) \equiv (p \vee \neg p)$.

Answer: This is true.

5. $(p \iff \neg p) \equiv p$.

Answer: This is false as the equivalence does not hold when $p = T$.

6. Suppose there exists a student in the CS department who owns a computer, there are at least two students in the CSE department and Ram is one of them. Then, Ram owns a computer.

Answer: As there are at least two students in the CSE department. This may be true or false. However, this can also be interpreted as neither.

7. Let x be a variable and $f = x^2 + 4x - 5$ be a polynomial. Then, there exists a natural number a such that $f(a) = 0$.

Answer: This is true. Take $a = 1$.

8. Let x be a variable and $P(x), Q(x)$ be predicates. Then, $\neg\exists x(P(x) \implies \neg Q(x)) \equiv \forall x(P(x) \wedge Q(x))$.

Answer: This is true. We know from DeMorgan's laws for quantifiers that $\neg\exists x(P(x) \implies \neg Q(x)) \equiv \forall x\neg(P(x) \implies \neg Q(x))$. This is further equivalent to $\forall x\neg(\neg P(x) \vee \neg Q(x))$. On applying DeMorgan's law of propositional logic, we get that this is equivalent to $\forall x(P(x) \wedge Q(x))$.

9. Suppose every student in IIIT Delhi has a phone. Then, Annirudh also has a phone.

Answer: This may be true or false (also interpreted as neither) as it is not given to us whether Annirudh is a student of IIIT Delhi.

10. Let x be a variable and $P(x), Q(x)$ be predicates. Then, $\neg\forall x(P(x) \wedge \neg Q(x)) \equiv \exists x(\neg P(x) \vee Q(x))$.

Answer: This is true. It can be shown using an argument similar to the one used in the 8th part above.

Question 1.3. Let p, q, r be propositional variables. Do the following by using only the algebra of propositional logic. You must mention the rule of algebra at every step.

1. Convert $(\neg(p \oplus \neg q) \iff r)$ to an equivalent CNF.

2. Convert $\neg(p \wedge (q \oplus r))$ to an equivalent DNF.

3. Convert $(p \implies \neg q) \oplus (q \iff (r \implies \neg r))$ to an equivalent conjunctive form (need not be a CNF).

4. Convert $(p \iff \neg q) \implies (q \oplus (p \wedge \neg r))$ to an equivalent disjunctive form (need not be a DNF).

Answer hint: This can easily be done by following the procedure taught in the course. In the last two parts, we don't have to obtain the normal forms, i.e., CNF and DNF.

Question 1.4. Let u be an arbitrary propositional formula defined using propositional variables $p_1, \dots, p_n$.

1. (3 marks). Let v be the DNF obtained from the truth table of u by following the procedure taught in the class. Prove that $u \equiv v$.

Answer hint: Already done in the class.

2. (6 marks). Let y be the CNF obtained from the truth table of u by following the procedure taught in the class. Prove that $u \equiv y$.

Answer hint: The proof of this follows the same line as the proof of the previous part. Recall the procedure to create a CNF from the truth table of a proposition. We mark all the false row in the truth table of the proposition u and then we create an OR-term for every marked row as follows: Pick a marked row arbitrarily. If the propositional variable p_i in this row is false, include p_i in the OR-term, otherwise include $\neg p_i$ in the OR-term. The final CNF y is the AND of each of these OR-terms. Now, we show that for every marked row, if we substitute the assignments of the $p_1, \dots, p_n$ in this row inside y , we get false. If we substitute the assignment of $p_1, \dots, p_n$ in any true row (or unmarked row) in the truth table of u inside the CNF y , we get true. This proves that u and y are equivalent.

3. (1 mark). Why don't we have to prove the above-mentioned equivalence if v, y are constructed by following the procedures for constructing CNFs and DNFs using the algebra of propositional logic?

Answer: Because when we construct CNFs or DNFs using the rule of algebra of propositions, we replace a proposition with an equivalent proposition at every step.

Question 1.5. 1. (5 marks) Let $p_1, p_2, p_3, p_4, p_5, p_6$ be propositional variables. Let u be the propositional formula defined as

$$u := (\neg p_1 \vee p_2) \wedge (\neg p_3 \vee p_4) \wedge (\neg p_2 \vee p_3) \wedge (\neg p_5 \vee p_6) \wedge (\neg p_6 \vee p_1) \wedge (\neg p_4 \vee p_5).$$

How many $a_1, a_2, a_3, a_4, a_5, a_6$ exist such that every a_i is either true or false, and when we substitute p_i with a_i in u for every $1 \leq i \leq 6$, we get true? Justify your answer.

(Hint: Can commutativity help to nicely restructure u ?)

Answer: By using commutativity of $\wedge$, we get that

$$u \equiv (\neg p_1 \vee p_2) \wedge (\neg p_2 \vee p_3) \wedge (\neg p_3 \vee p_4) \wedge (\neg p_4 \vee p_5) \wedge (\neg p_5 \vee p_6) \wedge (\neg p_6 \vee p_1).$$

We want to count the number of assignments to propositional variables $p_1, \dots, p_6$ which make u true. As u is a conjunctive form, it becomes true if each of its OR-term becomes true. We know every $p_1, \dots, p_6$ takes either T or F. Let's start with p_1 . We first set $p_1 = T$. Then, the OR-term $(\neg p_6 \vee p_1)$ becomes true, but, to make the OR-term $(\neg p_1 \vee p_2)$ true, p_2 should be set to T. Now, the OR-term $(\neg p_2 \vee p_3)$ becomes T if and only if $p_3 = T$. Using similar argument; to make $(\neg p_3 \vee p_4)$ true, p_4 should be T; to make $(\neg p_4 \vee p_5)$ true, p_5 should be true; and to make $(\neg p_5 \vee p_6)$ true, p_6 should be T. Thus, the conclusion is that if $p_1 = T$ and we want to make u also T, then every variable in $p_2, \dots, p_6$ should be set to true.

Now, let's see what happens when we assign $p_1 = F$. Then, the OR-term $(\neg p_1 \vee p_2)$ becomes T, but to make $(\neg p_6 \vee p_1)$ true, we must set $p_6 = F$. Now, to make $(\neg p_5 \vee p_6)$ true, we must set $p_5 = F$. Similarly, to make u true, we must set all variables false.

Thus, there are two satisfying assignments of u , namely $p_1 = p_2 = p_3 = p_4 = p_5 = p_6 = T$ and $p_1 = p_2 = p_3 = p_4 = p_5 = p_6 = F$.

2. (5 marks). Let u be an arbitrary CNF and v be an arbitrary DNF in propositional variables $p_1, \dots, p_n$.
 - (a) Does there exist an OR-term w_1 containing variable p_i for every $1 \leq i \leq n$, with or without negation such that w_1 is not present in u and $u \wedge w_1 \equiv u$? Justify your answer.

Answer: No. This is because u and $u \wedge w_1$ are equivalent if and only if they have the same truth tables. This implies that the rows which are false in the truth tables of $u \wedge w_1$ and u are the same. Hence, they should correspond to the same OR-terms. This implies that w_1 should be present in u .

Remark: This can also be proved using the algebra of propositional logic.

- (b) Does there exist an AND-term w_2 containing variable p_i for every $1 \leq i \leq n$ with or without negation such that w_2 is not present in v and $v \vee w_2 \equiv v$?

Answer. No. The argument is same as the previous part.

We state some definitions for the last question.

Definition 1 (Covering portion of an AND-term). Let $p_1, \dots, p_n$ be propositional variables and u be an AND-term in some variables $p_1, \dots, p_k$ for some natural number $k < n$. Let v be the proposition

$$v := (u \wedge \neg p_{k+1} \wedge \dots \wedge \neg p_{n-1} \wedge \neg p_n) \vee (u \wedge \neg p_{k+1} \wedge \dots \wedge \neg p_{n-1} \wedge p_n) \vee \dots \vee (u \wedge p_{k+1} \wedge \dots \wedge p_{n-1} \wedge p_n),$$

i.e., v is the OR of the AND-terms constructed using u by introducing every remaining variable $p_{k+1}, \dots, p_n$ not appearing in u , once with negation and other time without negation. In other words, v is the DNF equivalent to the AND-term u . Then, v is said to be the covering portion of u .

For example, $(p \wedge q \wedge \neg r) \vee (p \wedge q \wedge r)$ is the covering portion of the AND-term $(p \wedge q)$.

Definition 2 (Cover of a disjunctive form). Let u be a disjunctive form and v be a DNF in propositional variables $p_1, \dots, p_n$. Then, v is said to be the cover of u if every AND-term of u has a covering portion in v .

For example, let $u = (p \wedge q) \vee (p \wedge r)$ and $v = (p \wedge q \wedge \neg r) \vee (p \wedge q \wedge r) \vee (p \wedge \neg q \wedge r)$. Then, it is easy to verify that v is a cover of u .

Similarly, we can define the covering portion of an OR-term and the cover of a conjunctive form.

Question 1.6 (Uniqueness of DNFs). 1. (2.5 marks). Let u_1 be an AND-term in the propositional variables $p_1, \dots, p_n$. Let v_1 be the covering portion of u_1 . Prove that $u_1 \equiv v_1$.

(Hint: Use the rules of algebra of propositional logic).

Answer: First assume that u_1 contains all variables $p_1, \dots, p_n$. Then, by Definition 1, v_1 and u_1 are the same, and hence equivalent. Now suppose u_1 contains variables $p_1, \dots, p_k$ for some natural number $k < n$. Then, by Definition 1,

$$v_1 := (u_1 \wedge \neg p_{k+1} \wedge \dots \wedge \neg p_{n-1} \wedge \neg p_n) \vee (u_1 \wedge \neg p_{k+1} \wedge \dots \wedge \neg p_{n-1} \wedge p_n) \vee \dots \vee (u_1 \wedge p_{k+1} \wedge \dots \wedge p_{n-1} \wedge p_n). \quad (1)$$

Let's first work with two AND-terms of v_1 . Let $w_1 := (u_1 \wedge \neg p_{k+1} \wedge \dots \wedge \neg p_{n-1})$. By associativity of AND, we get

$$(u_1 \wedge \neg p_{k+1} \wedge \dots \wedge \neg p_{n-1} \wedge \neg p_n) \vee (u_1 \wedge \neg p_{k+1} \wedge \dots \wedge \neg p_{n-1} \wedge p_n) \equiv (w_1 \wedge \neg p_n) \vee (w_1 \wedge p_n).$$

Now, by using the distributivity law, we get that

$$(w_1 \wedge \neg p_n) \vee (w_1 \wedge p_n) \equiv w_1 \wedge (\neg p_n \vee p_n).$$

By using the validity for OR, we get $\neg p_n \vee p_n \equiv T$. Now, using the identity for AND, we get $w_1 \wedge T \equiv w_1$. Hence,

$$(u_1 \wedge \neg p_{k+1} \wedge \dots \wedge \neg p_{n-1} \wedge \neg p_n) \vee (u_1 \wedge \neg p_{k+1} \wedge \dots \wedge \neg p_{n-1} \wedge p_n) \equiv (w_1 \wedge T) \equiv w_1.$$

By using this argument on other consecutive pairs AND-terms of v_1 as above, we get that

$$v_1 \equiv (u_1 \wedge \neg p_{k+1} \wedge \dots \wedge \neg p_{n-1}) \vee (u_1 \wedge \neg p_{k+1} \wedge \dots \wedge \neg p_{n-1}) \vee \dots \vee (u_1 \wedge p_{k+1} \wedge \dots \wedge p_{n-1}). \quad (2)$$

Now, using the same argument as above that we used to deal with the first two AND-terms of v_1 given in Equation (1), we get from Equation (2) that $(u_1 \wedge \neg p_{k+1} \wedge \dots \wedge \neg p_{n-1}) \vee (u_1 \wedge \neg p_{k+1} \wedge \dots \wedge \neg p_{n-1})$ is equivalent to $(u_1 \wedge \neg p_{k+1} \wedge \dots \wedge \neg p_{n-2}) \vee (u_1 \wedge \neg p_{k+1} \wedge \dots \wedge \neg p_{n-1})$. Now, we work with other consecutive AND-terms of v_1 in Equation (2) and get that

$$v_1 \equiv (u_1 \wedge \neg p_{k+1} \wedge \dots \wedge \neg p_{n-2}) \vee (u_1 \wedge \neg p_{k+1} \wedge \dots \wedge \neg p_{n-2}) \vee \dots \vee (u_1 \wedge p_{k+1} \wedge \dots \wedge p_{n-2}).$$

Like this, we continue until we reach at the propositional variable p_k , and every time we remove variables from p_{k+1} to p_n one by one from all the and-terms of v_1 . Ultimately, we get that

$$v_1 \equiv u_1.$$

Remark. Formally, this is done using proof by induction. We will see this proof method later.

2. (2.5 marks). Let u_2 be a disjunctive form in propositional variables $p_1, \dots, p_n$. Let v_2 be the cover of u_2 . Prove that $u_2 \equiv v_2$.

(Hint: take help from the first part of this question and use the definition of a disjunctive form).

Answer. As v_2 is the cover of the disjunctive form u_2 , we know from Definition 2 that for every AND-term u_1 in u_2 , there exists a covering portion of u_1 in the DNF v_2 . Let's first understand how to prove the equivalence of u_2 and v_2 first with an example, to understand the main idea.

Let $u_2 = (p \wedge q) \vee (p \wedge r)$ and $v_2 = (p \wedge q \wedge \neg r) \vee (p \wedge q \wedge r) \vee (p \wedge \neg q \wedge r)$. Then, v_2 is a cover of u_2 . We want to prove that $u_2 \equiv v_2$. Of course, in this example, we can use the truth table. But let's see how to do this using the algebra of propositional logic. Let's take the AND-terms of u_2 one by one. We first take $(p \wedge q)$. Its covering portion in v_2 is $(p \wedge q \wedge r) \vee (p \wedge q \wedge \neg r)$. Now, we consider the AND-term $(p \wedge r)$ of u_2 . Its covering portion in v_2 is $(p \wedge q \wedge r) \vee (p \wedge \neg q \wedge r)$. Note that the AND-term $(p \wedge q \wedge r)$ is present in both these covering portions. But it appears once in the DNF v_2 . We first duplicate this using the idempotence rule of OR. In particular, $(p \wedge q \wedge r) \equiv (p \wedge q \wedge r) \vee (p \wedge q \wedge r)$. Then,

$$v_2 \equiv (p \wedge q \wedge \neg r) \vee (p \wedge q \wedge r) \vee (p \wedge q \wedge r) \vee (p \wedge \neg q \wedge r)$$

Now, the first two AND-terms in the above equation correspond to the covering portion of $(p \wedge q)$ and the last two AND-terms of the above equation correspond to $(p \wedge r)$. We know from the first part that $(p \wedge q) \equiv (p \wedge q \wedge \neg r) \vee (p \wedge q \wedge r)$ and $(p \wedge r) \equiv (p \wedge q \wedge r) \vee (p \wedge \neg q \wedge r)$. This shows that $u_2 \equiv v_2$.

Now, let's come back to the general case. For every AND-term w of the DNF v_2 , we check how many AND-terms of u_2 have w in their covering portion. We use the idempotence rule of OR to create those many copies of w in the DNF v_2 . After this, we use commutativity of OR on the new v_2 to first collect the covering portion of the first AND-term of u_2 , then the second covering portion of u_2 , and so on. Finally, we use the result in the first part of this question to argue that $v_2 \equiv u_2$.

3. (2.5 marks). Let u_3, u_4 be disjunctive forms in the propositional variable $p_1, \dots, p_n$. Let v_3, v_4 be the covers of u_3, u_4 , respectively. Suppose $u_3 \equiv u_4$. Prove that v_3 and v_4 are the same DNFs.

(Hint: Take help from the second part of this question and the second part of Question 1.5).

Answer. We are given that v_3 and v_4 are the covers of u_3 and u_4 . Then, from the second part of this question, we know that $v_3 \equiv u_3$ and $v_4 \equiv u_4$. We are given that $u_3 \equiv u_4$, which implies that $v_3 \equiv v_4$. We want to show that v_3 and v_4 are the same DNFs, i.e., both have the same AND-term. Let w be an arbitrary AND-term of v_3 . Consider $v_4 \vee w$. As v_3 is a DNF, we know from the second part of Question 1.5 that w should be contained in v_3 . As w was an arbitrary AND-term of v_3 , we get that entire v_3 is preset in v_4 . Now, using a similar argument, we get that entire v_4 is contained in v_3 . Hence, v_3 and v_4 are the same DNFs.

4. (2.5 marks). Let w be an arbitrary proposition in variables $p_1, \dots, p_n$. Prove without using the truth table of w that there exists a unique DNF that is equivalent to w .

(Hint: What will you do if you get two different equivalent disjunctive forms from the third-last step of the procedure to construct DNF using the algebra of propositional logic? Does any part of this question help?)

Answer. While following the procedure to convert a propositional formula into a DNF using only the rules of propositional logic, at the second last step, we always get a disjunctive form equivalent to the original propositional formula. Now, if there are two or more equivalent disjunctive forms obtained at the second-last step of this procedure, by the third part of this question, the DNFs obtained for each of these disjunctive form in the last step of the procedure will be the same.

Remark. Similar to Question 1.6, we can also show using the same ideas that every proposition has a *unique* equivalent CNF.